

Hurudaya Narasimhan

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Education

KREA University <i>BSc. (Hons.) Mathematics and Computer Science</i>	August 2024 – Present Tada, India
Chennai Mathematical Institute <i>BSc. (Hons.) Mathematics and Computer Science</i>	August 2022 – July 2024 Chennai, India

Work Experience

Signal Encoding Algorithms & Antenna Design Consulting <i>VDesign Technologies</i>	March 2023 – August 2024 Puducherry, India
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Optimized encoding algorithms and implemented Viterbi decoding - achieved 40% reduction in signal noise. Worked on a beam-steering Ku-band patch antenna for lightweight UAVs. Streamlined onboarding processes for new engineers.

Faculty of Mathematics <i>Math Circle Chennai</i>	September 2024 – Present Chennai, India
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Designed curricula to teach advanced mathematics to 100+ middle school students. Developed hands-on activities to teach cryptography, modular arithmetic, Markov Chains etc.

Mathematics Tutor <i>Independent</i>	October 2023 – Present Online
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Helped gifted students achieve competitive success in mathematical olympiads. Students qualified to IMO, INMO/IMOTC and EGMO.

Formal Verification Consultant <i>Lending Labs/Kuliza</i>	May 2026 – Present Remote
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Worked with leaders in Loan-Origination-Software to help prove invariant properties and show that test cases are reduced when constructing new business rules.

Research & Projects

Novel Topological Anti-Cheat for Online Multi-Player FPS games (active)
Developed unique topological method to identify *likely* cheaters in FPS games by modeling player crosshair position on 2-spheres, fitting von Mises-Fisher distributions, and clustering anomalies with HDBSCAN.

Machine Learning for Collatz Conjecture Bounds (active)
The Collatz conjecture asks whether repeatedly applying $n \mapsto n/2$ (if even) or $n \mapsto 3n + 1$ (if odd) always reaches 1. I am currently training a neural network to predict stopping times from binary representations.

Formal Verification in Lean and Isabelle
Taught myself dependent type theory by automating proofs of Olympiad-style number theory problems. Also worked on formalizing the Erdos-Szekeres conjecture under a professor.

Reinforcement Learning for Game AI

Got bored; implemented Q-learning and reward shaping for Snake and Poker agents; experimented with exploration-exploitation trade-offs and convergence optimization.

Internships

Data Analytics Intern

2024

Riverbridge Investment Advisors

Chennai, India

Automated data analytics pipelines; analyzed financial data and business valuations for an Indian private equity firm.

Teaching Assistant

2023

Epsilon India Summer Camp

Chennai, India

TA'd at a mathematics camp for gifted students. Introduced non-Euclidean geometry, finite automata, and Diophantine equations. Delivered lecture on proof writing.

Awards & Honors

JEE Mains: Math 99.8th percentile, Overall 96.47th percentile

2022

National Testing Agency (India)

SAT: 800 Math, 730 English

2021

CollegeBoard

National Robo-Soccer Champion and Overall Team Winner

2018

SP Robotics Works (SPARC)

Technical Skills

Programming Languages: Python, Haskell, R, SQL, Arduino

Machine Learning: PyTorch/TensorFlow, scikit-learn, Reinforcement Learning

Formal Methods: Lean, Isabelle

Data Analysis: pandas, NumPy, matplotlib, Statistical Analysis